

1 Solution — GIAM 1.7

Problem 3

1.1 Problem

What relation on the numbers from 1 to 10 does the given set of ordered pairs represent?

$$\{(1, 1), (1, 2), \dots, (1, 10), (2, 2), \dots, (2, 10), \dots, (9, 9), (9, 10), (10, 10)\}$$

1.2 Answer

the "less than or equal to" relation, $\leq$.

A pair (a, b) is in the set exactly when $a \leq b$.

1.3 Reading the Pattern

Look at how the pairs are organised by first coordinate:

first coord. a	the second coords. b that appear	condition
1	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	$b \geq 1$
2	2, 3, 4, 5, 6, 7, 8, 9, 10	$b \geq 2$
3	3, 4, 5, 6, 7, 8, 9, 10	$b \geq 3$
$\vdots$	$\vdots$	$\vdots$
9	9, 10	$b \geq 9$
10	10	$b \geq 10$

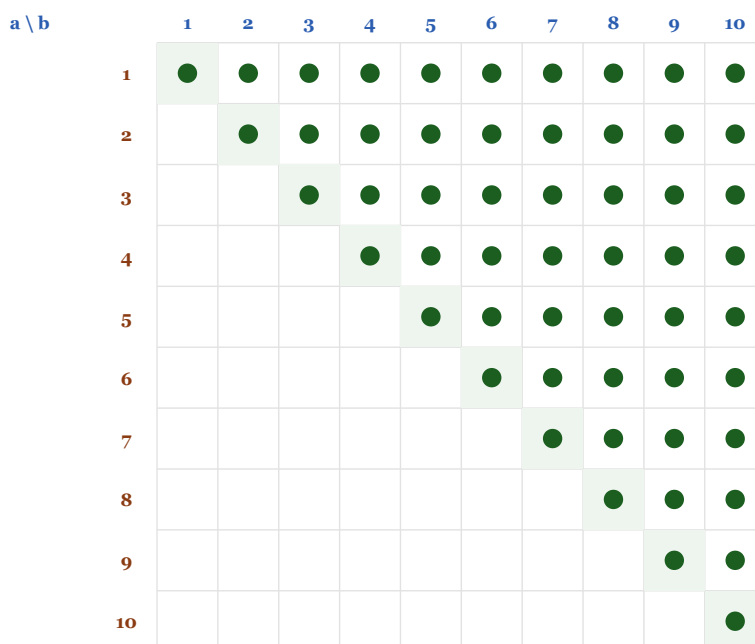
Table 1.1

For each first coordinate a , the partners b are precisely those with $b \geq a$ — i.e. $a \leq b$. So the relation is $\leq$.

1.4 The Relation as a Grid

Every cell on or above the main diagonal is filled — the entire upper triangle:

A dot sits at (a, b) exactly when $a \leq b$. Every cell on or above the diagonal is filled — the relation is $\leq$. The shaded diagonal ($a = b$) is included, so it is " $\leq$ " and not the strict " $<$ ".



Each row a runs from column a rightward: row a has $(11 - a)$ dots. The whole upper triangle is solid.

$$55 \text{ pairs} = 10 + 9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 = (10 \cdot 11)/2.$$

Figure 1.1: A 10 by 10 grid, rows $a = 1..10$ and columns $b = 1..10$, with a dot in cell (a,b) exactly when a is less than or equal to b . The entire region on and above the main diagonal is solid. Row 1 has all 10 dots, row 2 has 9 (columns 2..10), and so on down to row 10 with a single dot at (10,10). The shaded diagonal $a=b$ is included, marking that this is the non-strict relation $\leq$ rather than strict $<$. Total $55 = 10+9+8+7+6+5+4+3+2+1 = (10 \text{ times } 11)/2$.

1.5 Remark — The Diagonal Tells $\leq$ Apart From $<$

The decisive detail is that the diagonal pairs $(1, 1), (2, 2), \dots, (10, 10)$ are **present**. That makes the relation **reflexive** ($a \leq a$ always holds), which is exactly what separates $\leq$ from the strict $<$. The section's own example listed $<$ on $\{1, \dots, 5\}$ *without* any (k, k) pair. So the single distinction between this problem and that example is the diagonal: include it and you have $\leq$; drop it and you have $<$.

1.6 Remark — Contrast With the Divisibility Relation (Problem 1)

Both this relation and divisibility (Problem 1) live in the upper triangle of the grid, because $a \leq b$ and $a \mid b$ both force $a \leq b$. But $\leq$ fills the triangle *completely* (55 pairs), while divisibility keeps only the cells where a actually divides b (27 pairs). In fact divisibility is a **sub-relation** of $\leq$: whenever $a \mid b$ we also have $a \leq b$, so every divisibility pair is a $\leq$ pair, but not conversely (e.g. $(2, 3)$ is a $\leq$ pair but not a divisibility pair). Divisibility is a *partial* order — some pairs like 4 and 6 are incomparable — whereas $\leq$ is a **total** (linear) order: any two of the numbers are comparable, which is precisely why its grid has no gaps in the triangle.

1.7 Remark — Counting the Pairs: A Triangular Number

The number of pairs is

$$10 + 9 + 8 + \dots + 2 + 1 = \frac{10 \cdot 11}{2} = 55,$$

the **tenth triangular number**. This is no coincidence: counting pairs (a, b) with $1 \leq a \leq b \leq 10$ is the same as *choosing 2 of the numbers $1, \dots, 10$ with repetition allowed*, which is $\binom{10+1}{2} = \binom{11}{2} = 55$ (the stars-and-bars count from Section 1.4). Equivalently, of the $10^2 = 100$ cells in the full grid, the 10 diagonal cells sit on the line $a = b$, and the remaining 90 split evenly into 45 below and 45 above; the $\leq$ relation takes the 45 strictly-above plus the 10 diagonal = 55.

1.8 Remark — The Three Order Properties Are Visible in the Grid

That this relation is a (total) order can be read straight off the picture:

- **Reflexive** — the diagonal is fully shaded ($a \leq a$).
- **Antisymmetric** — no cell (a, b) with $a \neq b$ has its mirror (b, a) also filled; all off-diagonal dots lie strictly on one side. (If $a \leq b$ and $b \leq a$ then $a = b$.)
- **Transitive** — if (a, b) and (b, c) are dots then so is (a, c) ($a \leq b$ and $b \leq c \Rightarrow a \leq c$).

Reflexive + antisymmetric + transitive is the definition of a partial order; the extra fact that *every* pair is comparable (the triangle is completely filled) upgrades $\leq$ to a **total order**. This is the cleanest, most familiar order relation, and it is the yardstick against which the more subtle divisibility order of Problem 1 is measured.